

Anh Ho

678-644-5003 | hoanh@stanford.edu | [linkedin.com/in/anhho123](https://www.linkedin.com/in/anhho123) | github.com/onhole

EDUCATION

Stanford University

B.S. in Electrical Engineering, QuestBridge and Vida Valiente Scholar, GPA: 3.8

Stanford, CA

Expected Jun 2028

Georgia Institute of Technology

Dual Enrollment in Mathematics, GPA: 4.0

Atlanta, GA

Aug 2024 – May 2025

EXPERIENCE

R&D Digital Signal Processing Intern

Keysight Technologies

Jun 2026 – Sep 2026

Colorado Springs, CO

- Developed a breakthrough solution in a six-year-standing project: achieved live signal separation on a single twisted-pair, full-duplex 1000BASE-T1 transmission line without using directional couplers.
- Recovered 100% of live Ethernet packets sent every $300\mu\text{s}$ – 1ms over a 3 – 5ms capture from receivers and transmitters.
- Scaled the separation ability from 66.67 to 750MBd symbol rate while maintaining 100% symbol alignment with IEEE 802.3bw and 802.3bp's Test Mode 4.
- Optimized FIR filter implementation from $O(N^2)$ to $O(N \log N)$ time complexity.
- Designed a PHY-agnostic solution compatible across IEEE 802.3bw and IEEE 802.3bp automotive Ethernet standards.

PCB Designer

Stanford Sensor Mesh Network (SMESH)

Feb 2026 – Present

Stanford, CA

- Develop sensor nodes to measure environmental conditions to inform prescribed burn decisions.
- Designed a breakout PCB for the Heltec V3 and connected sensors and power interfaces to significantly decrease bring-up cost and time. [[GitHub Repository](#)]

PROJECTS

Variable Power Supply | *Analog circuit design, oscilloscope*

- Designed and built an adjustable DC bench supply.
- Characterized regulation and ripple across loads (100Ω – $10k\Omega$), achieving 17.3 — 17.9V output with $\leq 1.9\text{mVpp}$ ripple.
- Validated operation up to $\sim 173\text{mA}$ load current (3.0W delivered at 17.3V into 100Ω) and documented results in a lab report (schematics + measurements).

Bare-Metal Bring-Up | *STM32, Cortex-M4, C/C++*

- Authored custom startup code and linker script, implementing fully bare-metal firmware with direct register manipulation and deterministic timing.
- Wrote SysTick and TIM drivers for PWM generation and timing control without HAL/RTOS.
- Debugged at register level using GDB and OpenOCD.

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB

Hardware: STM32, ARM Cortex-M4, Heltec V3 (ESP32)

Tools: Altium Designer, KiCad, Keysight Infiniium, LTspice, GDB, OpenOCD, Git/GitHub

CLASSES

Engineering: Circuits I (EE101A), Physics Electromagnetism (Physics 43), Physics Mechanics (Physics 41)

Computer Science: Programming Abstraction (CS106B), Harvard's Introduction to CS, Cybersecurity, and AI (online)